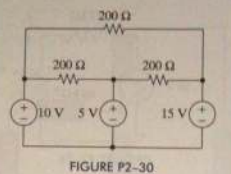


2-30 In Figure P2-30:



- Assign a voltage and current variable to every element.
- Use KVL to find the voltage across each resistor.
- Use Ohm's law to find the current through each resistor.
- Use KCL to find the current through each voltage source.

2-33 In Figure P2-33  $i_s = 0.5$  mA. Find the value of  $R$ .

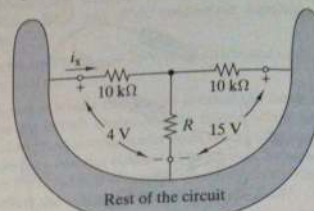


FIGURE P2-33

2-36 Find the equivalent resistance  $R_{EQ}$  in Figure P2-36.

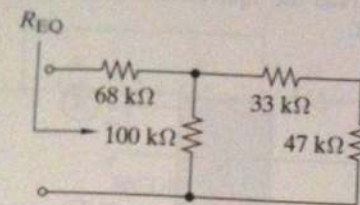


FIGURE P2-36

2-37 Find the equivalent resistance if  $R_{EQ}$  in Figure P2-37.

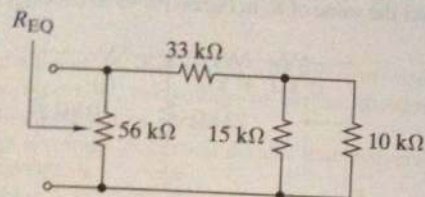


FIGURE P2-37

2-54 Use voltage division in Figure P2-54 to find  $v_x$ .

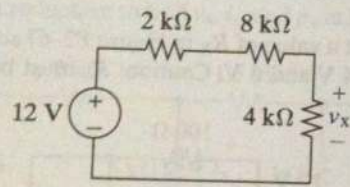


FIGURE P2-54

2-58 Find  $i_x$ ,  $i_y$ , and  $i_z$  in Figure P2-58.

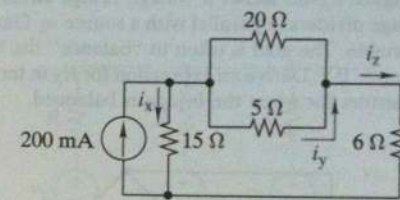


FIGURE P2-58

2-61 Find the range of values of  $v_O$  in Figure P2-61.

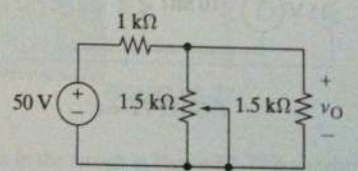


FIGURE P2-61

2-68 Use circuit reduction to find  $v_x$  and  $i_x$  in Figure P2-68.

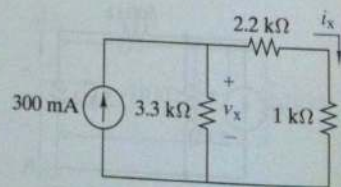


FIGURE P2-68

2-72 Use circuit reduction to find  $v_x$  and  $i_x$  in Figure P2-72.

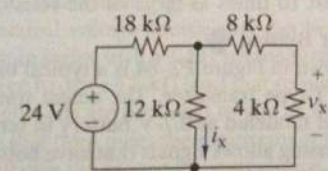


FIGURE P2-72

2-75 Use source transformations in Figure P2-75 to relate  $v_O$  to  $v_1$ ,  $v_2$ , and  $v_3$ .

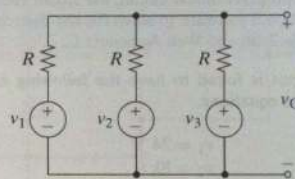


FIGURE P2-75